# Introduction

## Delivering computing to young learners

*Oxford International Primary and Lower Secondary Computing* is a complete syllabus for computing education for ages 5–14 (Years 1–9). By following the program of learning set out in this series, teachers can feel reassured that their students have access to the computing skills and understanding that they need for their future education.

Find out more at:
**www.oxfordprimary.com/computing**.

## Structure of the book

This book is divided into six chapters, for Year 4 (ages 8–9).

1. **The nature of technology:** Introduction to microprocessors and how they help us at work and at home
2. **Digital literacy:** Finding information safely using a web search
3. **Computational thinking:** Using variables and conditional structures in a program
4. **Programming:** Making programs with different types of input and output
5. **Multimedia:** Changing how a document looks
6. **Numbers and data:** Using a spreadsheet to process data

## What you will find in each unit

- Introduction: An offline activity and a class discussion help students to start thinking about the topic.
- Lessons: Six lessons guide students through activity-based learning.
- Check what you know: A test and activities allow you to measure students' progress.

## What you will find in the lessons

Although each lesson is unique, they have common features: learning outcomes for each lesson are set out at the start; learning content delivers skills and develops understanding.



Every lesson involves a learning activity for the students.



Activities to extend students who are able to do more.



Questions check students' understanding of the lesson.

## Additional features

You will also find these features throughout the book:



The word cloud builds vocabulary by identifying key terms from the unit.



Suggestions for creative and artistic work.



Extra tasks that can be taken outside the classroom and into the home.



Advice on using computers responsibly in life.

Glossary Key terms are identified in the text and defined in the glossary at the end.

## Assessing student achievement

The final pages in each unit give an opportunity to assess student achievement.

- Developing: This acknowledges the achievement of students who find the content challenging but have made progress.
- Secure: Students have reached the level set out in the programme for their age group. Most should reach this level.
- Extended: This recognises the achievement of students who have developed above-average skills and understanding.

Questions and activities are colour-coded according to achievement level. Self-evaluation advice helps students to check their own progress.

## Software to use

We recommend Scratch for writing programs at this age. For other lessons, teachers can use any suitable software, for example: Microsoft Office; Google Drive software; LibreOffice; any web browser.

## Source files

You will see this symbol on some of the pages.

This means that there are extra files you can access to help with the learning activities. For example, Scratch programming files and downloadable images.

To access the files, click 'Download resources' at:
**www.oxfordprimary.com/computing**.

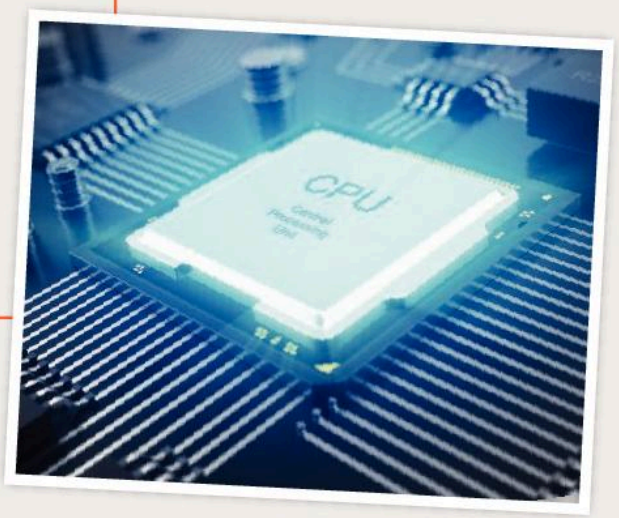
## Teacher's Guides

For more on these topics, look at the Teacher's Guide that accompanies this book.

# 1 The nature of technology: Computers around us

## You will learn

- that computer power improves how devices such as TVs and cars work
- how computers and technology improve the way people work
- about computer storage and why it is important.




Computer power is used to improve the way that devices work at home, at work and at school. TVs, cars, mobile devices, smartphones and fridges are all powered by computers. Computers are changing the way we live.



## Talk about...

If all computers disappeared tomorrow...

- what would you miss about computers in your home life?
- what would you miss about computers at school?





**Learning outcomes:** Describe how computers may be used in the world of work; Identify a range of modern devices which contain computer processors (for example, embedded processors); Describe what storage is and why it is important

## Class activity

Make a list of the types of computers you use at home and at school. For example, do you use a tablet computer? Do you use different types of computers to do different things?

microprocessor
robot sensors
storage drive
data file back-up file
flash drive



## Digital citizen of the future

Computers can improve the way we learn, work and enjoy our spare time. However, not everyone can afford to buy a computer. Some charities collect computers that are no longer used. These computers are given to people who cannot afford to buy their own. Would you donate a computer you no longer use?

## Did you know?

In 2015, scientists at Michigan University created Michigan Micro Mote, the world's smallest computer. The Mote measures 2mm × 2mm × 4mm. That's smaller than a grain of rice! The Mote is being used in medical implants and in driverless cars.